

# How much detail does the comparison need?

Two raw-data columns → test preservation and compactness → select the smallest qualifying representation

## 01 RAW VALUES

| Benign | Malignant |
|--------|-----------|
| 0.2699 | 1.0950    |
| 0.1852 | 0.5435    |
| 0.2773 | 0.7456    |
| 0.1839 | 0.4956    |
| 0.1563 | 0.7572    |
| 0.2636 | 0.3345    |

Six rows shown; all 569 values used.

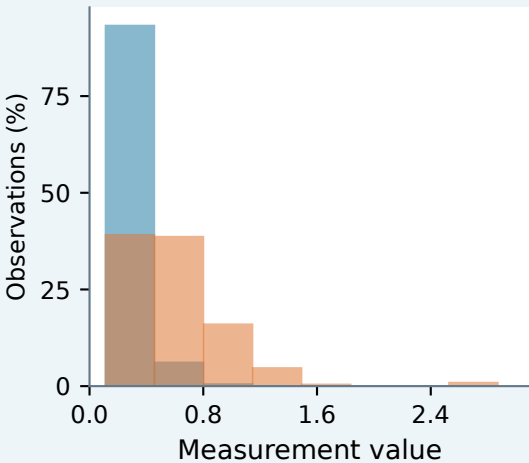
## 02 SCREEN 15 CHART FAMILIES

Match chart purpose to the question.  
7 eligible families → 21 display settings  
Exclude 8 families for other tasks

**Preserve the comparison:  $E \leq \text{tolerance}$**

Score = reduction in values (%)  
Floor at 0; reject with 0\*. Highest score first.

## 03 ONE DISPLAYED CHOICE



**Histogram / 96.8**

Here, a histogram is displayed: 18 of 569 values retained. A binned heatmap has equal numerical merit.

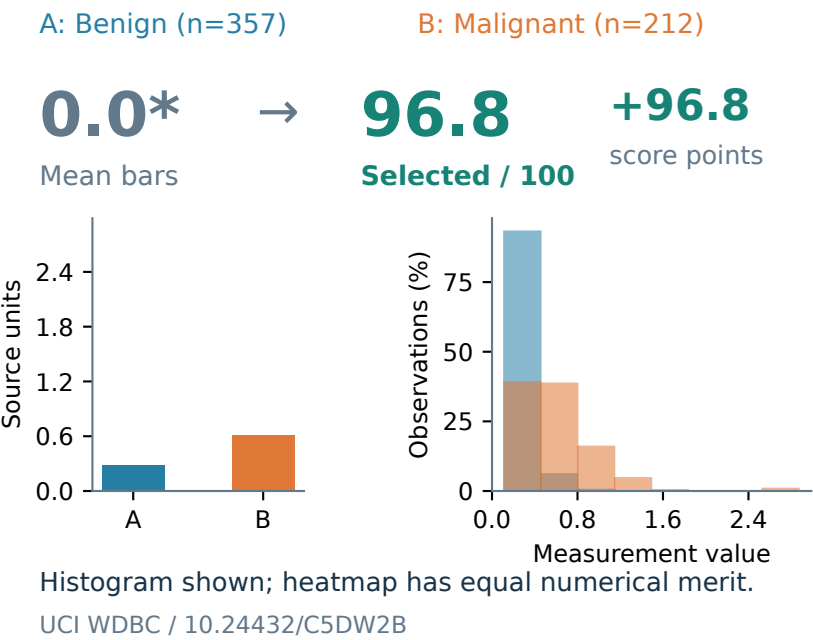
Primary audit: 61 mean, 29 five-number, 33 binned and 4 density selections across 127 comparisons.

The score describes eligible numerical simplification, not reader accuracy. Source: UCI WDBC / 10.24432/C5DW2B.

# Preserve the comparison with less detail

15 families screened / Seven eligible choices / One displayed recommendation

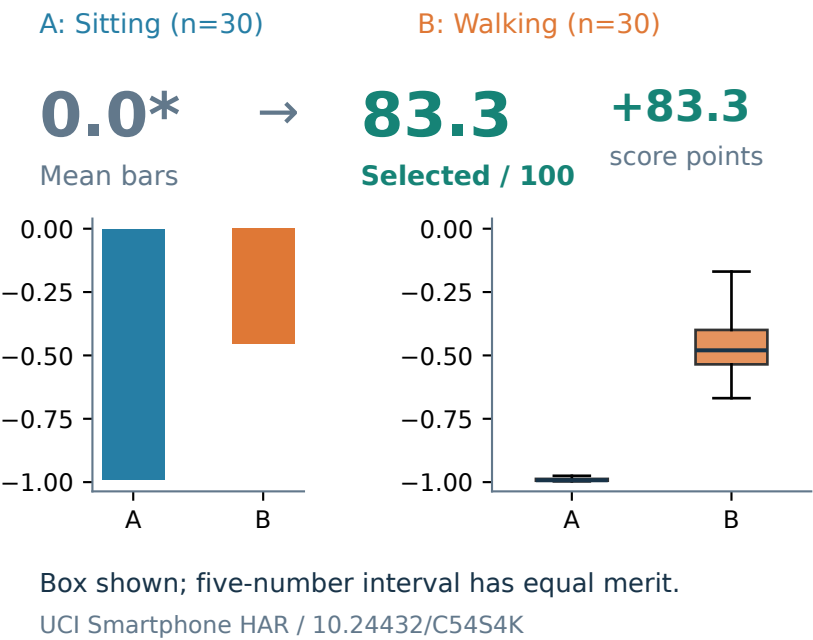
## a Breast tumour data / Radius standard error



|   | ORDER / ELIGIBLE CHART      | E / $\tau$ | SCORE | SIZE |
|---|-----------------------------|------------|-------|------|
| 1 | Histogram / 8 bins          | 0.348      | 96.8  | 18   |
| 2 | Heatmap / 8 bins            | 0.348      | 96.8  | 18   |
| 3 | Violin plot / 16 grid       | 0.610      | 94.0  | 34   |
| 4 | Dot plot / raw              | 0.000      | 0.0   | 569  |
| 5 | Bar chart                   | 2.018      | 0.0*  | 2    |
| 6 | Box plot                    | 5.699      | 0.0*  | 10   |
| 7 | Interval plot / five-number | 5.699      | 0.0*  | 10   |

Green: displayed choice.  $E / \tau \leq 1$  meets tolerance.  
Exact numerical ties use declared catalogue order.

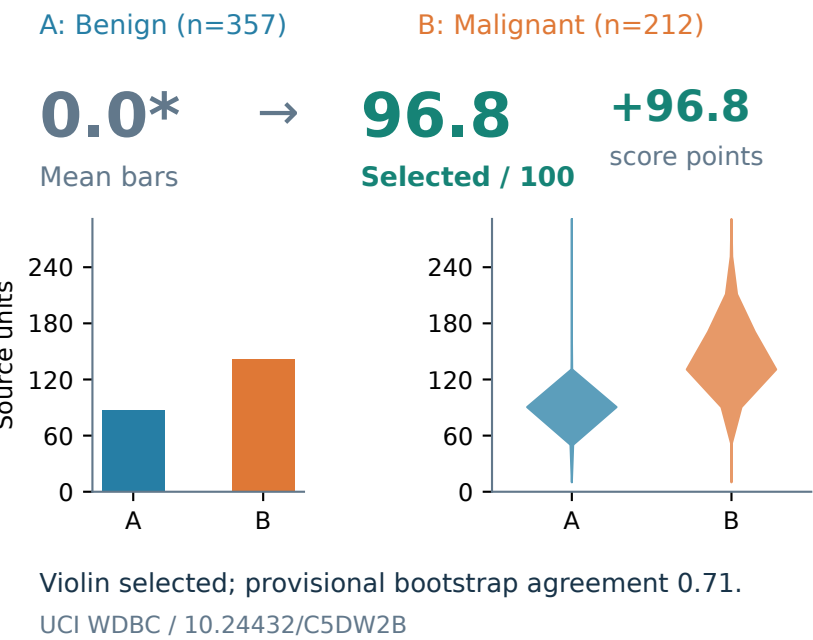
## b Smartphone activity data / Gyroscope-jerk variability, z axis



|   | ORDER / ELIGIBLE CHART      | E / $\tau$ | SCORE | SIZE |
|---|-----------------------------|------------|-------|------|
| 1 | Box plot                    | 0.164      | 83.3  | 10   |
| 2 | Interval plot / five-number | 0.164      | 83.3  | 10   |
| 3 | Histogram / 8 bins          | 0.547      | 70.0  | 18   |
| 4 | Heatmap / 8 bins            | 0.547      | 70.0  | 18   |
| 5 | Violin plot / 16 grid       | 0.369      | 43.3  | 34   |
| 6 | Dot plot / raw              | 0.000      | 0.0   | 60   |
| 7 | Bar chart                   | 1.445      | 0.0*  | 2    |

Green: displayed choice.  $E / \tau \leq 1$  meets tolerance.  
Exact numerical ties use declared catalogue order.

## c Breast tumour data / Worst perimeter



|   | ORDER / ELIGIBLE CHART      | E / $\tau$ | SCORE | SIZE |
|---|-----------------------------|------------|-------|------|
| 1 | Violin plot / 8 grid        | 0.378      | 96.8  | 18   |
| 2 | Histogram / 8 bins          | 0.427      | 96.8  | 18   |
| 3 | Heatmap / 8 bins            | 0.427      | 96.8  | 18   |
| 4 | Dot plot / raw              | 0.000      | 0.0   | 569  |
| 5 | Box plot                    | 1.662      | 0.0*  | 10   |
| 6 | Interval plot / five-number | 1.662      | 0.0*  | 10   |
| 7 | Bar chart                   | 1.892      | 0.0*  | 2    |

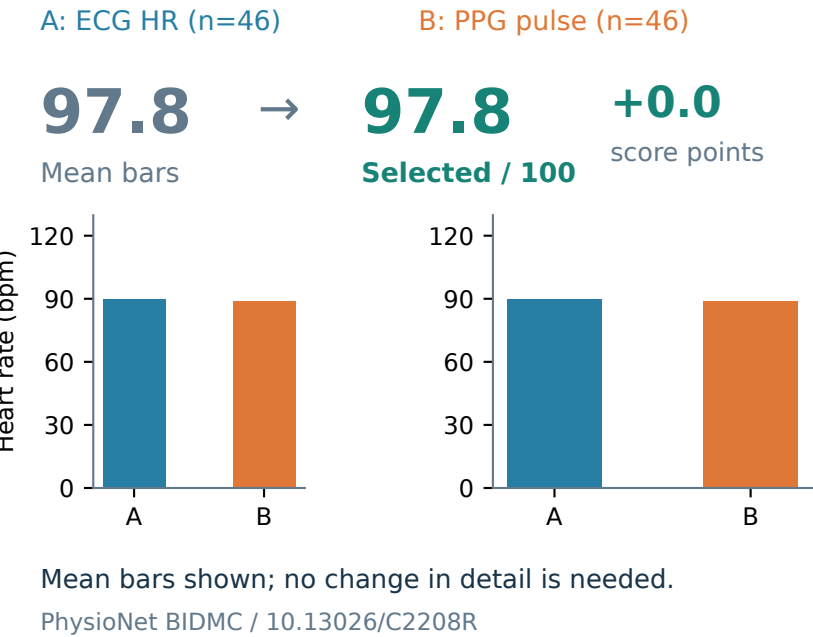
Green: displayed choice.  $E / \tau \leq 1$  meets tolerance.  
Exact numerical ties use declared catalogue order.

Score: percentage reduction in retained values, credited only when the comparison stays within tolerance.  
Raw dots = 0 (no reduction). 0\* = does not qualify. Equal positive scores: smaller discrepancy ranks first.  
All 15 families are documented in Table S3. Eight other-task families are excluded from these scored tables.

# Simplification has limits

No change may be needed / Long tails remain / Source extremes can require more detail

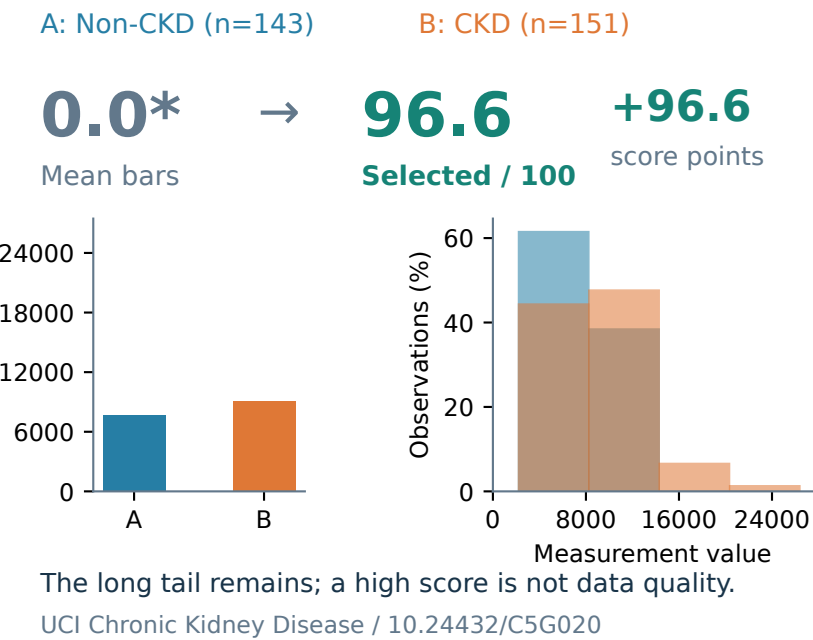
## a Physiological monitor data / Mean heart rate



| ORDER / ELIGIBLE CHART |                             | E / $\tau$ | SCORE | SIZE |
|------------------------|-----------------------------|------------|-------|------|
| 1                      | Bar chart                   | 0.573      | 97.8  | 2    |
| 2                      | Dot plot / means            | 0.573      | 97.8  | 2    |
| 3                      | Histogram / 4 bins          | 0.653      | 89.1  | 10   |
| 4                      | Heatmap / 4 bins            | 0.653      | 89.1  | 10   |
| 5                      | Box plot                    | 0.692      | 89.1  | 10   |
| 6                      | Interval plot / five-number | 0.692      | 89.1  | 10   |
| 7                      | Violin plot / 8 grid        | 0.528      | 80.4  | 18   |

Green: displayed choice.  $E / \tau \leq 1$  meets tolerance.  
Exact numerical ties use declared catalogue order.

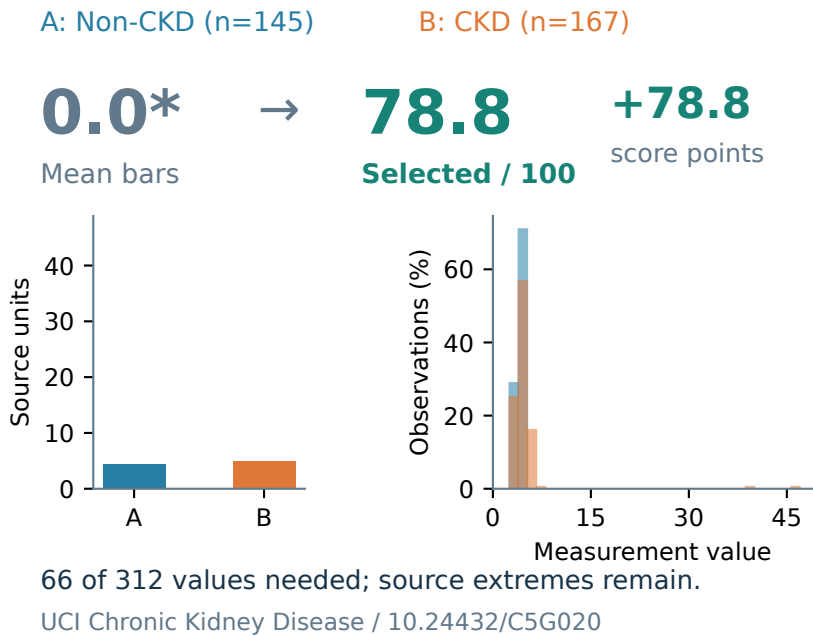
## b Kidney disease data / White blood cell count



| ORDER / ELIGIBLE CHART |                             | E / $\tau$ | SCORE | SIZE |
|------------------------|-----------------------------|------------|-------|------|
| 1                      | Histogram / 4 bins          | 0.954      | 96.6  | 10   |
| 2                      | Heatmap / 4 bins            | 0.954      | 96.6  | 10   |
| 3                      | Violin plot / 8 grid        | 0.697      | 93.9  | 18   |
| 4                      | Dot plot / raw              | 0.000      | 0.0   | 294  |
| 5                      | Bar chart                   | 1.254      | 0.0*  | 2    |
| 6                      | Box plot                    | 3.236      | 0.0*  | 10   |
| 7                      | Interval plot / five-number | 3.236      | 0.0*  | 10   |

Green: displayed choice.  $E / \tau \leq 1$  meets tolerance.  
Exact numerical ties use declared catalogue order.

## c Kidney disease data / Serum potassium



| ORDER / ELIGIBLE CHART |                             | E / $\tau$ | SCORE | SIZE |
|------------------------|-----------------------------|------------|-------|------|
| 1                      | Histogram / 32 bins         | 0.890      | 78.8  | 66   |
| 2                      | Heatmap / 32 bins           | 0.890      | 78.8  | 66   |
| 3                      | Dot plot / raw              | 0.000      | 0.0   | 312  |
| 4                      | Bar chart                   | 1.634      | 0.0*  | 2    |
| 5                      | Violin plot / 16 grid       | 1.662      | 0.0*  | 34   |
| 6                      | Box plot                    | 53.833     | 0.0*  | 10   |
| 7                      | Interval plot / five-number | 53.833     | 0.0*  | 10   |

Green: displayed choice.  $E / \tau \leq 1$  meets tolerance.  
Exact numerical ties use declared catalogue order.

Score: percentage reduction in retained values, credited only when the comparison stays within tolerance.  
Raw dots = 0 (no reduction). 0\* = does not qualify. Equal positive scores: smaller discrepancy ranks first.  
All 15 families are documented in Table S3. Eight other-task families are excluded from these scored tables.